Plotting Connections and Relationships Transcript

We will now move to another type of relationship that is understanding connections and relationships. You might appreciate all these are custom visuals and can be used for specific purposes. The general utility of these graphs or visualisations can be limited because the specific use cases as you might note the special visuals are reserved for specific occasions.

They are generally beaten as custom visualisations helping to portray a specific point. However, there are general visualisations which can be handy in most of the scenarios. We now move to the next category of graphs which are used for plotting connections and relationships.

These graphs are interesting in their own right. They help us explore the unknown. As I mentioned in earlier classes, one of the key steps of data visualisation is mining and exploration of data.

In this case, when you are encountered with a new data set in which you have variables but you don't know the relationship between the variables. In such a scenario when you don't know the relationship between the variables, you can use these plots to explore those variables and give insights. You can explore the relationship between variables and draw insights from these graphs and make a modelling exercise.

For example, what you are seeing on the slide is a palette of scatter plots. It is used to run correlation between all the variables in the data set and to understand how are they related to each other. For example, you can just run the command in Python and get a matrix of correlation plots and understand how two variables are related to each other.

This is very interesting to explore data where you have no clue about how variables are interacting with each other. This is called a scatter plot matrix because you have matrix of scatter plots. Let us say there are four variables in the data set A, B, C, D. You are running correlation between A versus A, A versus B, B versus C, C versus D, I mean A versus B, A versus C, and A versus D. This is called a scatter plot matrix amongst all variables in the data set and helps us to uncover patterns.

We can also use the size of bubbles to indicate shape or indicate the size to get the quantitative effect and at a single snapshot, it can reveal the relationship between all the variables in the data set. You also have a heat map which can be produced in a similar fashion where you can see the correlations between the variables across the data set and understand whether the data set is composed of similar variables or distinct variables. This is one of the fundamental steps in doing exploratory data analysis where you are trying to understand relationship between the variables.

We now move to the next type of chart which explores plot relationships and connections. We now look at the radial diagram or the quad, radial network or the quad diagram. This diagram treats all the variables as on a circle and removes the restriction of dealing it on a X and Y basis. Just to compare, look at the scatter plot matrix where you have pair, you can compare only a pair of variables in at any given instance. The X and Y are fixed. As opposed to this, in a radial diagram, there is no restriction of one variable's connection with other.

You can simultaneously study the relationship of a variable with others removing the dimensionality problem. It is simply portrayed as a circle and the connections indicate the intensity of the relationship. This is very interesting to see how a group of objects are related to others, where is the relationship stronger or which direction the relationship is stronger.

Radial network also captures the imagination of the audience immediately which requires some amount of study as opposed to scatter plot which would require some amount of mental effort. Another plotting and connection relationship graphic is the network plot or the network diagram. Here, it might look complicated, but this is a very useful diagram to understand the network and relationship between different variables which are part of an hierarchy or a system.

You can understand the nodes and the connections which are driving the relationship in the network. For example, you can take the graphic that is given on your slide. You can see the central parts are highlighted in red followed by the next important parts and the periphery are intuitively coloured in lighter colours.

You can see which is the centre of the network and how strongly it is connected to the other parts in the network and you can draw such inferences. The algorithms which provide these visualisations try to group the objects based on the existing relationships providing a sense to the data which we are not able to visualise numerically. This will help us in understanding the trends and patterns that evolve over a period of time especially social network analysis or peer network analysis can be done using these network graphs can be in a very interesting case study.

So, these are advanced visualisations which help us to understand the structure of underlying data.

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